Supermarket Sales Data Analysis

Power BI ETL & Excel Pivot Tables Assignment

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2026

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1 Part 1 – Power BI ETL

The following steps document the full Extract, Transform, and Load (ETL) process performed in Power BI Desktop using Power Query Editor and DAX.

1.1 Step 1: Extract

- Opened Power BI Desktop and selected **Get Data** → **Excel Workbook**
- Navigated to `Supermarket Dataset.xlsx` and selected the **Orders** sheet
- Clicked **Transform Data** to open Power Query Editor before loading
- Inspected column headers, data types, and sample values to identify transformation needs
- Enabled Column Quality, Column Distribution, and Column Profile in the View ribbon for full data profiling

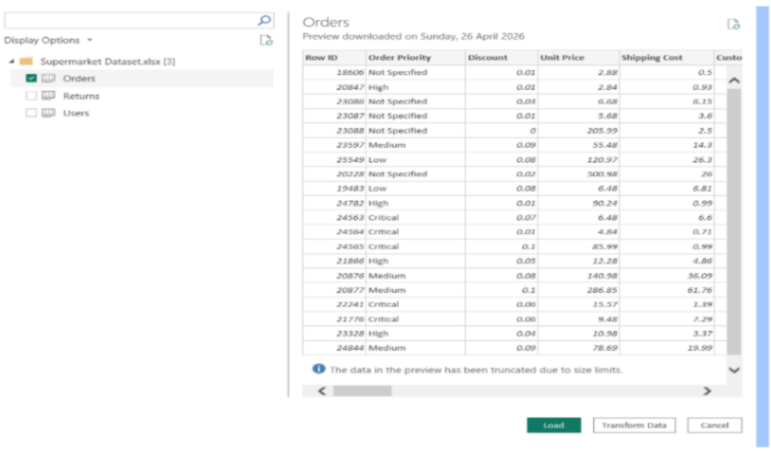


Figure 1: Power Query data profiling view

Columns identified as needing transformation:

Column	Issue Identified	Action Required
Order Date	Loaded as text/mixed type	Convert to Date type
Ship Date	Loaded as text/mixed type	Convert to Date type
Postal Code	Treated as number (loses leading zeros)	Change to Text type
Customer Name	Possible leading/trailing spaces	Trim whitespace
Product Category	Possible leading/trailing spaces	Trim whitespace
Region	Possible leading/trailing spaces	Trim whitespace
Sales	Nulls present	Remove null rows
Profit	Nulls present	Remove null rows

1.2 Step 2: Transform (Power Query)

2.1 Date Handling

- Selected the **Order Date** column → right-clicked → Change Type → Date
- Selected the **Ship Date** column → right-clicked → Change Type → Date
- Created a new custom column: **Shipping Days**

$$\text{Shipping Days} = [\text{Ship Date}] - [\text{Order Date}]$$

- Changed the Shipping Days column type to Whole Number

2.2 Derived Columns

- Created a new custom column: **Profit Margin %**

$$\text{Profit Margin \%} = ([\text{Profit}]/[\text{Sales}]) \times 100$$

- Created a conditional column: **Discount Band**

Condition	**Assigned Label**
Discount $= 0$	No Discount
$0 <$ Discount ≤ 0.05	Low
$0.05 <$ Discount ≤ 0.10	Medium
Discount > 0.10	High

2.3 Data Cleaning

- Filtered out rows where Sales or Profit is null
- Applied `Text.Trim` to Customer Name, Product Category, and Region columns

```
= Table.TransformColumns(Source, {{"Customer Name", Text.Trim},
  {"Product Category", Text.Trim}, {"Region", Text.Trim}})
```

- Changed Postal Code column type to Text

2.4 Aggregations (Group By)

- **Aggregation Table 1 – Regional Product Performance:** Group By: Region, Product Category; Sum of Sales (Total Sales), Sum of Profit (Total Profit). Saved as `agg_region_category`
- **Aggregation Table 2 – Shipping & Segment Counts:** Group By: Customer Segment, Ship Mode; Count of Order ID (Order Count). Saved as `agg_segment_shipmode`

1.3 Step 3: Load & Model

- Clicked **Close & Apply** to load cleaned tables into Power BI data model

DAX Date Table

`DateTable = CALENDAR(MIN(Orders[Order Date]), MAX(Orders[Order Date]))`

Helper columns:

- `Year = YEAR(DateTable[Date])`
- `Month = MONTH(DateTable[Date])`
- `Month Name = FORMAT(DateTable[Date], "MMMM")`
- `Quarter = "Q" & QUARTER(DateTable[Date])`

Created relationship between `DateTable[Date]` and `Orders[Order Date]`. Marked DateTable as official Date Table.

1.4 Step 4: Measures (DAX)

Measure Name	**DAX Formula**
Total Sales	`Total Sales = SUM(Orders[Sales])`
Total Profit	`Total Profit = SUM(Orders[Profit])`
Avg Profit Margin	`Avg Profit Margin = AVERAGE(Orders[Profit Margin %])`
YoY Sales Growth	`YoY Sales Growth = DIVIDE([Total Sales] - CALCULATE([Total Sales],`
Profit per Order	`Profit per Order = DIVIDE([Total Profit], DISTINCTCOUNT(Orders[Orde`

1.5 Step 5: Report Visuals

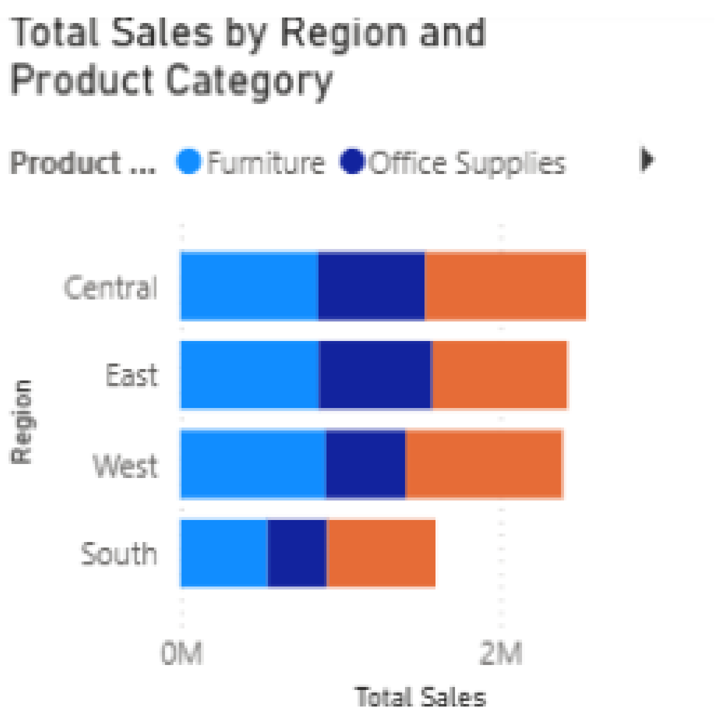


Figure 2: Sales by Region and Product Category – Stacked Bar Chart

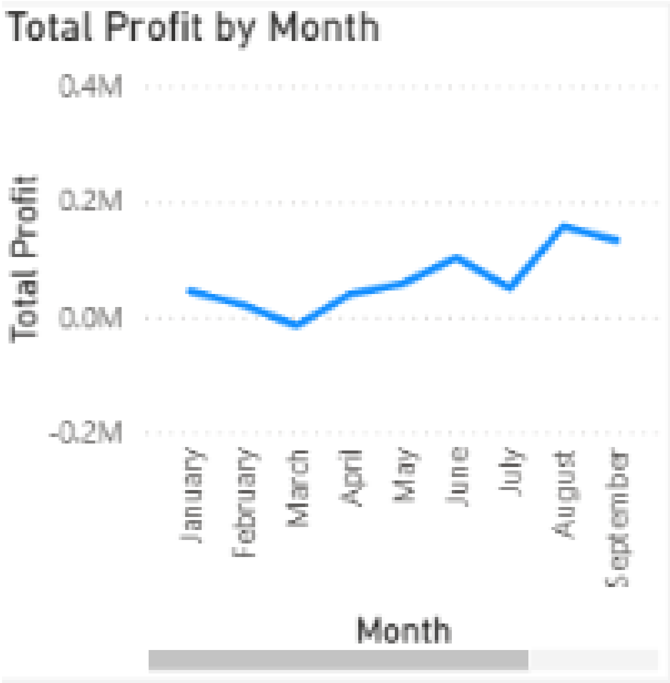


Figure 3: Profit Trend Over Time – Line Chart

Power BI Insight

Insight: The West region consistently records the highest Total Sales across all product categories, particularly in Technology. However, when the Profit Margin % measure is applied, the West also shows elevated discount usage in the Furniture category, which suppresses its profitability despite high revenue. This suggests that the West region is volume-driven but margin-constrained in Furniture — a key finding for pricing strategy decisions.

2 Part 2 – Excel Pivot Table Analysis

2.1 Pivot Table 1 – Sales & Profit by Region

- **Sheet tab:** PT1 – Region Sales Profit
- Rows: Region; Values: Sum of Sales, Sum of Profit
- Numbers formatted as Currency (USD, 2 decimals)
- Pivot Chart: Clustered Column chart comparing Sales and Profit side-by-side

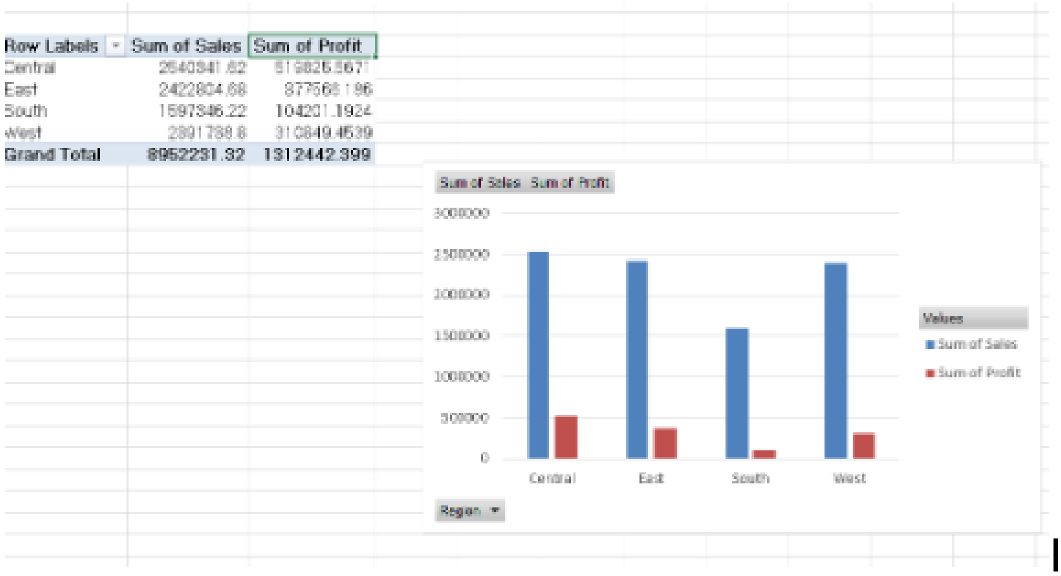


Figure 4: Sales vs Profit by Region

2.2 Pivot Table 2 – Profitability by Customer Segment and Ship Mode

- **Sheet tab:** PT2 – Segment Ship Margin
- Rows: Customer Segment; Columns: Ship Mode
- Values: Average Profit Margin % (calculated field: Profit / Sales)
- Conditional formatting: Green fill for top 25%, Red fill for bottom 25%

Sum of Profit Margin Pct	Column Labels			
Row Labels	Delivery Truck	Express Air	Regular Air	Grand Total
Consumer	6%	20%	15%	11%
Corporate	7%	24%	22%	15%
Home Office	5%	19%	17%	13%
Small Business	11%	20%	25%	19%
Grand Total	**7%**	**21%**	**20%**	**15%**

Figure 5: Profitability by Segment and Ship Mode

2.3 Pivot Table 3 – Top 10 Products by Profit

- **Sheet tab:** PT3 – Top 10 Products
- Rows: Product Name; Values: Sum of Profit
- Sorted descending; Top 10 filter applied
- Slicer inserted for Product Category

Row Labels	Sum of Profit
Electrix Halogen Magnifier Lamp	11123.9045
GE 48" Fluorescent Tube, Cool White Energy Saver, 34 Watts, 30/Box	9491.8147
Global Leather and Oak Executive Chair, Black	11834.73122
Global Troy Executive Leather Low-Back Tilter	79509.3922
Lifetime Advantage Folding Chairs, 4/Carton	12554.9879
Luxo Professional Fluorescent Magnifier Lamp with Clamp-Mount Base	15528.2143
Office Star - Mid Back Dual function Ergonomic High Back Chair with 2-Way Adjustable Arms	10638.17914
SAFCO Arco Folding Chair	11325.7232
SAFCO Folding Chair Trolley	14455.3322
Tenex Contemporary Contur Chairmats for Low and Medium Pile Carpet, Computer, 39" x 49"	11291.6019
Grand Total	**187753.8813**

Product Category
Furniture
Office Supplies
Technology

Figure 6: Top 10 Products by Profit with Category Slicer

2.4 Pivot Table 4 – Monthly Sales Trend

- **Sheet tab:** PT4 – Monthly Sales Trend
- Rows: Order Date (grouped by Months); Values: Sum of Sales, Count of Order ID
- Pivot Chart: Combo Chart (Line for Sales, Column for Order Count)

2.5 Pivot Table 5 – Discount Analysis

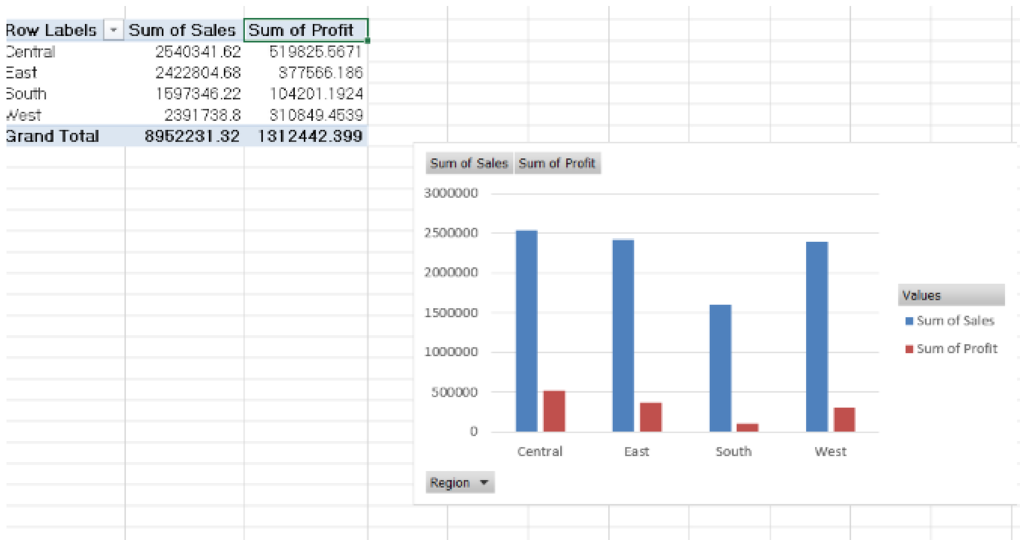
- **Sheet tab:** PT5 – Discount Analysis
- Created Discount Band column using nested IF:

```
=IF([@Discount]=0,"No Discount",IF([@Discount]<=0.05,"Low",
  IF([@Discount]<=0.10,"Medium","High")))
```

- Rows: Discount Band; Values: Average Profit Margin %, Sum of Sales

2.6 Pivot Table 6 – Geographic Drill-down

- **Sheet tab:** PT6 – Regional Drilldown
- Rows: Region (outer), State or Province (inner)
- Values: Sum of Profit
- Collapsible rows to identify most profitable states per region



Row Labels	Sum of Sales	Sum of Profit
Central	2540341.62	519825.5671
East	2422804.68	377566.186
South	1597346.22	104201.1924
West	2391738.8	310849.4539
Grand Total	8952231.32	1312442.399

Figure 7: Geographic Drill-down by Region and State

Excel Pivot Table Insight

Insight: The Discount Analysis pivot table (PT5) reveals a clear inverse relationship between discount level and profitability. Products sold with High discounts (>10%) show an average profit margin significantly below those sold with No Discount or Low discounts. Furthermore, the Monthly Sales Trend pivot (PT4) shows that peak order volume months do not always correspond to peak revenue months — suggesting that promotional periods drive order counts up while simultaneously compressing margins through discounting.

3 Submission Summary

Deliverable	File / Location	Status
Power BI file	Advanced ETL Model.pbix	Includes all transformations, Date Table, 5 DA
Excel workbook	Supermarket Dataset.xlsx	6 pivot tables on separate labelled worksheet ta
Documentation	ETL_Task2_Documentation.docx	This document — steps, transformations, and i

End of Documentation